Petra Dobric

Personal presentation: What do I expect to learn?

I am a student on the final year of Master's programme in Biotechnology in Medicine at the University of Rijeka. I have a bachelor's degree in Biotechnology and Drug Development that I also acquired at the same university.

I have very basic knowledge in bioinformatics which I really want to expand. I believe that bioinformatics is a great field to be in right now because it offers a lot of opportunities. One of them is being able to work from your computer remotely which gives you more flexibility, unlike the other potential career paths that I could take such as working in a laboratory.

I chose this course to expand my knowledge, which is at a very basic level, and to hopefully gain more hands-on experience because I did not have a chance to do that at my home university. My offer in courses like this is very slim, therefore I was very interested in partaking in this subject. I expect to get a better understanding of how the biological data is analysed using various computational tools, which tools are used for which step of the analysis, and apply some of them myself in the practical.

Assay of two attended seminars

<u>Seminar 1: Exploring the role of transposable elements in adaptation and lifespan</u>

This seminar, held by Miriam Merenciano, gives us an overview of transposable elements (TEs) and their role primarily using *Drosophila* as a model organism. Transposable elements, discovered by Barbara Mc Clintock in maze, are DNA fragments that can move around in the genome. They are conserved throughout species and occupy around 50% of human genome. They were originally referred as selfish mutagens and junk DNA, however further research revealed that they are in symbiosis with their hosts as well as drive genome evolution.

A great model organism to study this change is *Drosophila* because of its short life span, easy maintenance in the laboratory and many adaptive changes in the genome. In *Drosophila melanogaster,* TE insertions like the *roo* element, have shown to provide alternative transcription sites for genes like the *Lime* gene, which has been linked to immune and metabolic responses. Experiments have proven that these TE insertions influence gene expression and directly increase the flies' chances of survival during infections. On the other hand, in *Drosophila suzukii* the Mrp4 TE insertion is connected to a stronger response to oxidative stress, which itself does not completely increase survival.

In addition, the sex gap longevity (SGL) was also explored in this seminar. Many animal species display different mortality patterns, where the heterogametic sex is usually the one with a shorter lifespan. This raises a belief that the Y chromosome may contribute to sex-specific aging often referred to as the “toxic Y effect”. The presented research indicates that TE expression increases with age and is observed in both sexes in the carcass. The said increase is specifically pronounced in the gonads, suggesting that age-related genomic instability is closely tied to the loss of control over transposable elements.

<u>Seminar 2: Spatial Transcriptomics</u>

In this seminar, held by Irepan Salvador-Martinez, we learned what is transcriptomics and some recent advances in this field. He showed us the evolution of methods that are most commonly used as well as explained the differences between them and the benefits of using each method.

We started of with bulk RNA sequencing which was developed in the early 2000’s, it is a method that gives us an average expression level of a gene across a large sample. This is good for determining a disease biomarker or studying broad level differences, but sometimes we need more detailed information, and averages are not very useful. This is where Single-cell RNA sequencing works better and increases the amount of information we can extract from a sample. By using single cell RNA sequencing, we get a better insight into individual cells, cellular heterogeneity and subpopulation expression. This method is more expensive and complex than bulk RNA sequencing, however it provides us with a bigger amount of information.

We were introduced to Spatial Transcriptomics, a technology that allows biologists to view the biology of single cells while retaining information on how they are distributed within a tissue, and two methods that are currently relevant for this field. Firstly, Xenium (10x), an imaging-based profiling technology which can map a large number of RNA targets at a subcellular resolution without requiring sequencing. It has been used to analyse lung and colon patterns to study gene expression pathways. The other technique that was mentioned is STAMP. It is a new, accessible platform that “stamps” cells in suspension onto imaging slides. It provides a low-cost, multimodal (RNA and protein) method for characterizing cell states, like those seen in human embryonic stem cells during gastrulation.

Final discussion: What I learned?

During this course I was introduced to many technologies currently used in bioinformatics that I wasn’t aware of beforehand, which gave me a better perspective of how this field of biology is applied in experimental research and data analysis. I was able to upgrade my R programming skills, while also revising some basics that I already knew. In addition, I gained insight into biological data interpretation and got to work in a group setting which improved my collaborative skills.